

AL-TAHS On-Prem Operations SOP

1) Purpose

This SOP defines how to run AL-TAHS safely in an on-prem (local) environment, prevent data loss, and recover quickly from failures.

2) Scope

Applies to:

- Application host PC (Windows)
- Backend/API service
- PostgreSQL database
- Local backup and restore-test automation
- LAN/VPN user access

3) System Profile

- Deployment type: On-prem local system
- Database: PostgreSQL (localhost:5432)
- App access: LAN/VPN users only
- Backup root: C:\altas-backups\pg
- Daily backup task: `altas Daily Postgres Backup`
- Weekly restore-test task: `altas Weekly Restore Test`

4) Daily Start Checklist (Operator)

- 1 Confirm host PC is powered on and stable.
- 2 Confirm PostgreSQL service is running.
- 3 Confirm backend server is running and login page is reachable.
- 4 Verify users can log in from LAN/VPN.
- 5 Check disk free space on system drive and backup drive (minimum 20% free recommended).

5) Daily End-of-Day Data Protection

- 1 Ensure scheduled task `altas Daily Postgres Backup` is enabled.
- 2 Confirm a fresh backup exists under C:\altas-backups\pg\daily\.
- 3 Confirm success state file updated:
 - C:\altas-backups\pg\state\latest_success.json
- 1 If failure occurred, inspect:
 - C:\altas-backups\pg\state\latest_failure.json
 - Latest .log file in C:\altas-backups\pg\daily\

6) Weekly Backup Validity Proof (Mandatory)

- 1 Ensure scheduled task `altas Weekly Restore Test` is enabled.
- 2 Confirm latest restore-test success state:
 - `C:\altas-backups\pg\state\latest_restore_test_success.json`
- 1 Check restore-test logs:
 - `C:\altas-backups\pg\restore-test\*.log`
- 1 If restore-test fails, treat as high priority and fix before business week starts.

7) Incident Response

A) App unavailable

- 1 Check backend process and restart if needed.
- 2 Check PostgreSQL service status.
- 3 Check firewall/network reachability (LAN/VPN).
- 4 Review backend logs and latest backup/restore-test states.

B) Data corruption or major error

- 1 Stop writes (pause app use).
- 2 Create emergency backup first.
- 3 Restore from most recent valid backup.
- 4 Validate core modules: POS, HR, Reports, Promotions, Sales SDC.
- 5 Resume service only after validation.

8) Restore Procedure (Standard)

- 1 Identify latest valid backup in `C:\altas-backups\pg\daily\`.
- 2 Run restore to target DB using approved restore command/script.
- 3 Verify:
 - Users can log in
 - Recent transactions visible
 - Key tables present
- 1 Record restore date/time, operator, and backup file used.

9) Security and Access Rules

- Keep system on private LAN/VPN only (no public exposure).
- Use role-based user accounts only; never share credentials.
- Rotate admin credentials periodically.
- Restrict backup folder access to authorized operators.
- Do not disable backup/restore-test tasks.

10) Change Control

Before updates (schema, dependencies, major features):

- 1 Take manual pre-change backup.

- 2 Apply change in maintenance window.
- 3 Run smoke test.
- 4 Confirm backup and restore-test still pass.
- 5 Document change and rollback path.

11) Monthly Audit Checklist

- 1 Verify scheduled tasks still enabled and successful.
- 2 Confirm restore-test logs show clean pass.
- 3 Review storage growth and retention behavior.
- 4 Test user access from at least one branch/LAN client.
- 5 Update SOP contacts and escalation list if changed.

12) Ownership and Escalation

- System Owner: _____
- Technical Operator: _____
- Backup Reviewer: _____
- Escalation Contact: _____

If backup fails for more than 1 day or restore-test fails once, escalate immediately and do not delay corrective action.